# *INTELLIGENT HOSTEL RESOURCE ALLOCATION AND RESIDENT MANAGEMENT SYSTEM USING SPRING BOOT AND MONGODB*


Name: Rajvardhan Gahlaut

Registration Number: 23BEC0081

Course: B.Tech Electronics and Communication Engineering

Institution: VIT Vellore

# ABSTRACT

Hostel management is an important aspect of educational institutions, requiring efficient handling of student accommodation, room allocation, visitor records, complaint management, and hostel administration. Traditional manual methods are often inefficient, time-consuming, and prone to errors.

The Intelligent Hostel Resource Allocation and Resident Management System is a backend-based application developed using Spring Boot and MongoDB. The system provides a centralized platform for managing hostel-related activities through RESTful APIs. It enables administrators to maintain student records, allocate rooms automatically, manage wardens, track complaints, monitor occupancy, and maintain visitor logs. The project aims to improve operational efficiency, data consistency, and resource utilization within hostel environments.

# INTRODUCTION

Educational institutions with hostel facilities need an organized system to manage residents, rooms, staff, and administrative activities. Manual record-keeping systems often lead to data redundancy, inconsistencies, and difficulties in retrieving information.

The proposed Intelligent Hostel Resource Allocation and Resident Management System addresses these challenges by providing a digital solution for hostel administration. The system utilizes modern backend technologies and a NoSQL database to store and manage data efficiently.

The project focuses on creating a scalable and maintainable backend application that supports CRUD operations, database relationship mapping, automated room allocation, and complaint management. By leveraging Spring Boot and MongoDB, the system ensures flexibility, high performance, and ease of integration with future frontend applications.

## PROBLEM STATEMENT

Managing hostel operations manually can result in several issues, including:

1. Difficulty in maintaining student records.
2. Inefficient room allocation processes.
3. Lack of centralized complaint management.
4. Poor tracking of visitor entries and exits.
5. Challenges in monitoring room occupancy.
6. Data duplication and inconsistency.
7. Time-consuming administrative procedures.

A digital hostel management solution is required to overcome these limitations and improve overall administrative efficiency.

## OBJECTIVES

1. To develop a backend-based Smart Hostel Management System using Spring Boot and MongoDB.

2. To manage student, hostel, room, warden, visitor, and complaint information efficiently.
3. To implement CRUD operations for all major entities in the system.
4. To establish One-to-One, One-to-Many, Many-to-One, and Many-to-Many relationships in MongoDB.
5. To automate room allocation based on room availability and capacity.
6. To maintain occupancy records for effective hostel resource utilization.
7. To provide a complaint management system for hostel residents.
8. To maintain visitor records for security and monitoring purposes.
9. To expose RESTful APIs that can be tested using Postman and documented using Swagger.
10. To demonstrate practical implementation of backend development concepts and database management techniques.

# TECHNOLOGY STACK

### Backend Framework

Spring Boot is used for developing RESTful APIs and handling business logic.

### Programming Language

Java is used as the primary programming language due to its robustness, portability, and enterprise-level support.

### Database

MongoDB is used as the NoSQL database for storing and managing hostel-related information.

### API Testing

Postman is used to test API endpoints and verify functionality.

### API Documentation

Swagger is used to generate interactive API documentation.

# <u>SYSTEM MODULES</u>

### 1. Student Management Module

This module manages student-related information.

Functions:

- Add student
- View student details
- Update student information
- Delete student records
- Search students

### 2. Hostel Management Module

This module maintains hostel information.

Functions:

- Create hostel records
- Update hostel details
- Delete hostel information
- View hostel details

### 3. Room Allocation Module

This module manages room assignments.

Functions:

- Allocate rooms
- Transfer students between rooms
- Track room occupancy
- Monitor room availability

**4. Warden Management Module**

This module manages hostel wardens.

Functions:

- Add wardens
- Update warden details
- Assign wardens to hostels
- View warden information

**5. Complaint Management Module**

This module handles student complaints.

Functions:

- Register complaints
- Update complaint status
- Resolve complaints
- View complaint history

### 6. Visitor Management Module

This module tracks visitor information.

Functions:

- Visitor entry registration
- Visitor exit registration
- Visitor history tracking
- Daily visitor monitoring

### 7. Analytics and Reporting Module

This module provides hostel statistics.

Functions:

- Total students
- Total hostels
- Total rooms
- Occupied rooms
- Vacant rooms
- Pending complaints
- Resolved complaints

# DATABASE DESIGN

The project uses MongoDB collections for storing data.

**Collections**

**Students**

- Student ID
- Name
- Email
- Phone Number
- Department
- Room ID

**Hostels**

- Hostel ID
- Hostel Name
- Hostel Type

**Rooms**

- Room ID
- Room Number

- Capacity
- Occupancy Status

**Wardens**

- Warden ID
- Name
- Contact Number

**Complaints**

- Complaint ID
- Student ID
- Complaint Description
- Status

**Visitors**

- Visitor ID
- Visitor Name
- Entry Time
- Exit Time

# DATABASE RELATIONSHIPS

**One-to-One Relationship**

Student ↔ Hostel ID Card

Each student is associated with a unique hostel identification card.

**One-to-Many Relationship**

Hostel → Rooms

One hostel contains multiple rooms.

Room → Students

One room can accommodate multiple students.

Student → Complaints

A student can register multiple complaints.

**Many-to-One Relationship**

Rooms → Warden

Multiple rooms can be supervised by a single warden.

**Many-to-Many Relationship**

Students ↔ Visitors

A student may receive multiple visitors, and visitors may visit different students over time.

## CONCLUSION

The Intelligent Hostel Resource Allocation and Resident Management System provides a modern and efficient solution for hostel administration. The project demonstrates the practical implementation of Spring Boot, MongoDB, REST APIs, and database relationship mapping. By automating hostel operations and centralizing information management, the system improves efficiency, reduces administrative workload, and enhances resource utilization.

The project successfully fulfills the objectives of managing students, rooms, wardens, complaints, and visitors while implementing CRUD operations and MongoDB relationships. It serves as a scalable foundation for future hostel management solutions and real-world backend application development.